

Notes for Model Reduction

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Balanced Truncation

There is a well-known equality:

$$\sigma_i(\mathcal{H})^2 = \lambda_i(W_o W_c) \quad (1)$$

where $\mathcal{H}$ represents system's Hankel matrix, W_o represents the observability Grammian, W_c represents controllability Grammian, σ_i are Hankel singular values and λ_i represents the eigenvalues.

Then the error bound is:

$$\|G_n - G_r\|_{H_\infty} \leq 2(\sigma_{r+1} + \dots + \sigma_n)$$

where $\sigma_{r+1}, \dots, \sigma_n$ are the neglected Hankel singular values.

The goal of balanced truncation is to find an invertible transformation $T \in \mathbb{C}^{n \times n}$, such that

$$TPT^* = \hat{P} = \Sigma$$

where Σ is diagonal matrix of the Hankel singular values of the system.

One efficient and numerically more robust implementation of the Balanced Truncation algorithm is the square-root (SR) method:

GPT

(1) You can interpret what each state represents

Each state corresponds to:

- a specific decay rate (ω_k),
- a specific direction in output space (y_k),
- a specific “shape” in time or frequency.

So you can say things like:

- “This state captures long-memory dynamics.”
- “This state only affects output channel 2.”
- “This state models fast transient behavior.”

That is impossible in a generic realization.

3. Interpretation in plain language

Algorithm 1 Square Root Method

1. Compute the Cholesky factorization of the Grammians:

$$P = Z_P Z_P^* \quad (2)$$

$$Q = Z_Q Z_Q^* \quad (3)$$

2. Then we write the “bridge” matrix, $Z_P^* Z_Q$ in the SVD coordinates:

$$Z_P^* Z_Q = \begin{pmatrix} U_1 & U_2 \end{pmatrix} \begin{pmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{pmatrix} \begin{pmatrix} V_1^* \\ V_2^* \end{pmatrix}$$

where $\begin{pmatrix} \Sigma_1 & \\ & \Sigma_2 \end{pmatrix}$ is the Hankel singular values due to equation (1).
 U_1, U_2, V_1, V_2 are the corresponding singular vector blocks.

Each state / basis function corresponds to:

- a temporal shape
 $\frac{1}{1-\omega_k z}$ (how the effect evolves in time),
 - and an input direction
 y_k (which combination of inputs feeds into that temporal shape).
- So you can literally say:
- “This state is driven mainly by input 1.”
 - “This state is driven by inputs 2 and 3 together.”
 - “This state responds to the difference between two inputs.”
 - “This state ignores input 4 completely.”

That interpretation is precise and invariant.

Appendix

Balanced truncation formula

First, let’s define the Cholesky factors of Grammians as in equations (2) and (3):

$$P = Z_P Z_P^*$$

$$Q = Z_Q Z_Q^*$$

where P is the controllability Grammian (How much input energy is required to excite each state direction):

$$P = \sum_{k=0}^{\infty} A^k B B^* (A^*)^k$$
$$P = A P A^* + B B^*$$

and Q is the observability Grammian (large value means state strongly affects output):

$$Q = \sum_{k=0}^{\infty} (A^*)^k C^* C A^k$$

$$Q = A^* Q A + C^* C$$

Let's further perform the SVD of $Z_P^* Z_Q$:

$$Z_P^* Z_Q = U \Sigma V^* = \begin{pmatrix} U_1 & U_2 \end{pmatrix} \begin{pmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{pmatrix} \begin{pmatrix} V_1^* \\ V_2^* \end{pmatrix}$$

If we define

$$W^* = \Sigma_1^{-1/2} V_1^* Z_Q^T$$

then

$$W = (Z_Q^T)^* V_1 \left(\Sigma_1^{-1/2} \right)$$

$$= \bar{Z}_Q V_1 \Sigma_1^{-1/2}$$

and

$$V^* = \Sigma_1^{-1/2} U_1^* Z_P^T$$

then we can achieve

$$W^* Q W = \Sigma_1$$

and

$$V^* P V = \Sigma_1$$

Proof

Balanced truncation needs two things on the kept subspace:

1. Bi-orthogonality:

$$W^* V = I_r$$

2. Balanced Gramians:

$$W^* Q W = \Sigma_1, V^* P V = \Sigma_1$$

First, let's prove 1:

$$W^* V = \Sigma_1^{-1/2} V_1^* Z_Q^T (V^*)^*$$

$$= \Sigma_1^{-1/2} V_1^* Z_Q^T \left(\Sigma_1^{-1/2} U_1^* Z_P^T \right)^*$$

$$= \Sigma_1^{-1/2} V_1^* Z_Q^T \bar{Z}_P U_1 \Sigma_1^{-1/2}$$

Key identity: $(A^T)^* = \bar{A}$. So we also have $Z_Q^T \bar{Z}_P = Z_Q^T (Z_P^T)^* = (Z_P^* Z_Q)^T$.
Therefore,

$$W^* V = \Sigma_1^{-1/2} V_1^* (Z_P^* Z_Q)^T U_1 \Sigma_1^{-1/2}$$

From SVD block identity $Z_P^* Z_Q V_1 = U_1 \Sigma_1$, we have:

$$V_1^T (Z_P^* Z_Q)^T = \Sigma_1 U_1^T$$

then

$$V_1^T (Z_P^* Z_Q)^T U_1 = \Sigma_1$$

then

$$W^* V = \Sigma_1^{-1/2} \Sigma_1 \Sigma_1^{-1/2} = I_r.$$

Now, let's prove 2:

$$W^* Q W = \left(\Sigma_1^{-1/2} V_1^* Z_Q^T \right) Z_Q Z_Q^* W \quad (4)$$

Since

$$W = (W^*)^* = \bar{Z}_Q V_1 \Sigma_1^{-1/2}$$

equation (4) becomes:

$$W^* Q W = \left(\Sigma_1^{-1/2} V_1^* Z_Q^T \right) Z_Q Z_Q^* \bar{Z}_Q V_1 \Sigma_1^{-1/2}$$

since $Z_P^* Z_Q V_1 = U_1 \Sigma_1$, we have

$$V_1^* Z_Q^* Z_P Z_P^* Z_Q V_1 = \Sigma_1^2 \quad (5)$$

$$V_1^* Z_Q^* P Z_Q V_1 = \Sigma_1^2 \quad (6)$$

Standard Method

If we define

$$W = Z_P U_1 \Sigma_1^{-1/2}$$

then

$$\begin{aligned} W^* Q W &= \left(\Sigma_1^{-1/2} U_1^* Z_P^* \right) Q Z_P U_1 \Sigma_1^{-1/2} \\ &= \Sigma_1^{-1/2} U_1^* (Z_P^* Q Z_P) U_1 \Sigma_1^{-1/2} \end{aligned}$$

but since $Q = Z_Q Z_Q^*$, we have

$$\begin{aligned} W^* Q W &= \Sigma_1^{-1/2} U_1^* Z_P^* Z_Q Z_Q^* Z_P U_1 \Sigma_1^{-1/2} \\ &= \end{aligned}$$

Then from the identity (5) and (6), we have

$$W^* Q W = \Sigma_1$$

And if we let

$$V = Z_Q V_1 \Sigma_1^{-1/2}$$

GPT BT Algorithm

Let

$$\begin{aligned} P &= Z_P Z_P^* \\ Q &= Z_Q Z_Q^* \end{aligned}$$

SVD:

$$Z_P^* Z_Q = \begin{pmatrix} U_1 & U_2 \end{pmatrix} \begin{pmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{pmatrix} \begin{pmatrix} V_1^* \\ V_2^* \end{pmatrix}$$

notice that we have something called SVD block identity:

$$Z_P^* Z_Q = U_1 \Sigma_1 V_1^* + U_2 \Sigma_2 V_2^*$$

then

$$Z_P^* Z_Q V_1 = U_1 \Sigma_1 \quad (7)$$

and

$$U_1 Z_P^* Z_Q = \Sigma_1 V_1^* \quad (8)$$

Then

$$W^* = \Sigma_1^{-1/2} U_1^* Z_P^*$$

$$W = Z_P U_1 \Sigma_1^{-1/2}$$

and

$$V = Z_Q V_1 \Sigma_1^{-1/2}$$

which means

$$V^* = \Sigma_1^{-1/2} V_1^* Z_Q^*$$

First,

$$\begin{aligned} W^* V &= \Sigma_1^{-1/2} U_1^* Z_P^* Z_Q V_1 \Sigma_1^{-1/2} \\ &= \Sigma_1^{-1/2} U_1^* U_1 \Sigma_1 \Sigma_1^{-1/2} \\ &= I_n \end{aligned}$$

Next,

$$\begin{aligned} W^* Q W &= \left(\Sigma_1^{-1/2} U_1^* Z_P^* \right) Q Z_P U_1 \Sigma_1^{-1/2} \\ &= \left(\Sigma_1^{-1/2} U_1^* Z_P^* \right) Z_Q Z_Q^* Z_P U_1 \Sigma_1^{-1/2} \\ &= \Sigma_1^{-1/2} U_1^* M M^* U_1 \Sigma_1^{-1/2} \end{aligned}$$

Now use the SVD: $M = U \Sigma V^*$. Then

$$\begin{aligned} M M^* &= U \Sigma V^* V \Sigma U^* \\ &= U \Sigma^2 U^* \end{aligned}$$

Therefore,

$$\begin{aligned}
U_1^* M M^* U_1 &= U_1^* U \Sigma^2 U^* U_1 \\
&= U_1^* \begin{pmatrix} U_1 & U_2 \end{pmatrix} \begin{pmatrix} \Sigma_1^2 & 0 \\ 0 & \Sigma_2^2 \end{pmatrix} \begin{pmatrix} U_1^* \\ U_2^* \end{pmatrix} U_1 \\
&= U_1^* (U_1 \Sigma_1^2 U_1^* + U_2 \Sigma_2^2 U_2^*) U_1 \\
&= \Sigma_1^2
\end{aligned}$$

Similarly,

$$V^* P V = \Sigma_1^{-1/2} V_1^* M^* M V_1 \Sigma_1^{-1/2}$$

Since $M^* M = V \Sigma^2 V^*$, thus

$$V_1^* M^* M V_1 = \Sigma_1^2$$

So

$$V^* P V = \Sigma_1$$

Then the reduced matrix is calculated by:

$$A_r = W^* A V, B_r = W^* B, C_r = C V$$

Sanity Checks

1. $U_1^* U_1 \approx I$
2. $V_1^* V_1 \approx I$, should be 10^{-10} to 10^{-6} .
3. $\|Z_P^* Z_Q V_1 - U_1 \Sigma_1\|$
4. $W^* V \approx I$
5. $W^* Q W \approx \Sigma_1, V^* P V \approx \Sigma_1$
6. Check if the transformation are ill conditioned.